

Applied Machine Learning

Benchmarking: Large-Scale Benchmarking & OpenML



OpenML

Learning goals

- Why Benchmarking?
- What is OpenML?

WHY BENCHMARKING

It is hard to evaluate methods using theoretical/mathematical analysis alone.

- Theoretical analysis typically requires strong assumptions (e.g., infinite sample size, no noise, ideal model class).
- Implementation details cannot be formally compared through purely theoretical/mathematical analysis (e.g., runtime, memory use).
- Effect of different data properties on algorithms cannot be explained theoretically, e.g., does XGBoost outperform an LM on data with only categorical features?
- Hyperparameter settings are often not part of the theoretical analysis.

THE STEPS OF BENCHMARKING

- ➊ Formulate a hypothesis (about algorithm performance or behavior)
- ➋ Define an appropriate performance metric (if not part of the hypothesis)
- ➌ Select or collect datasets to evaluate the hypothesis
- ➍ Define a resampling strategy
- ➎ Define hyperparameter search space and tuning strategy
- ➏ Execute the full cross-product of learners, datasets, and tuning configurations
- ➐ Analyze results and draw conclusions

THE PROBLEM WITH DATASETS

- No universally adopted standard dataset format
- No universally adopted dataset sharing platform
- Few platforms offer programmatic (API-based) access to data

THE IDEAL WORLD - FRICTIONLESS ML

Datasets

- Unified access to all datasets via unified API
- Possibility to share own data (mostly relevant in academic context)
- Discover and search datasets via their meta-data

Associated Objects

- **Tasks:** Predefined train/test splits for standardized evaluation
- **Runs:** Publicly shared results (algorithm, settings, performance)

OPENML - THE PROJECT

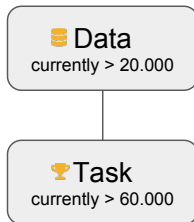
► OPENML

► OPENML is not only a data repository, it is a collaborative ML platform for sharing individual **components** involved in benchmark experiments.

Benchmarking: compare algorithms w.r.t. performance/runtime on datasets.

Goal: Reproducibility, transparency, and large-scale collaboration in ML.

OpenML relies on 4 **basic components** (just like benchmark experiments):



Supervised ML **tasks** contain:

- data + target feature
- performance measure
- resampling technique (e.g., 10-fold CV) and train-test splits

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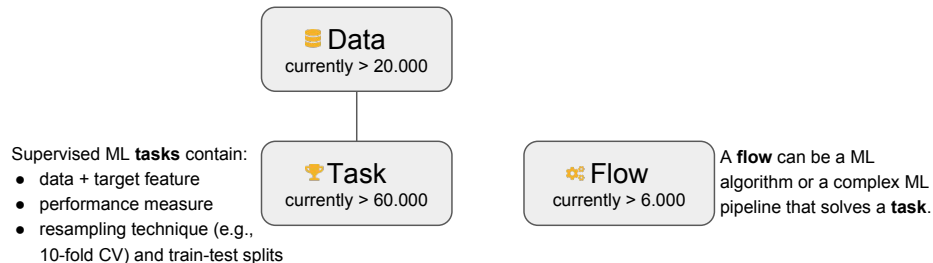
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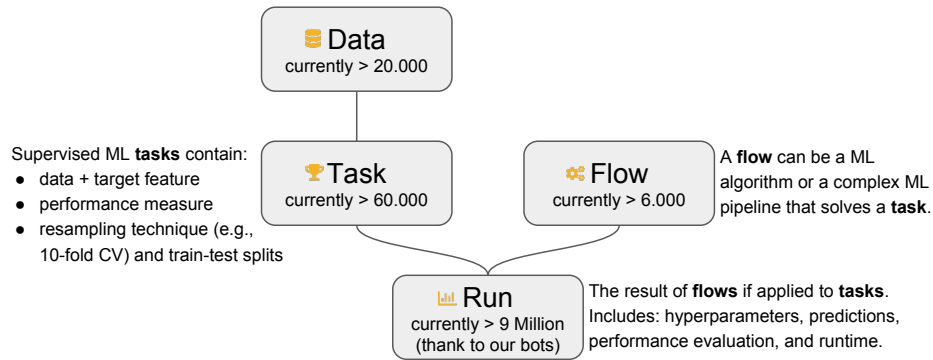
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Note: OpenML also supports **collections** of tasks and runs

OPENML DATASETS

- **Raw data** stored in standardized formats:
 - .arff (Attribute-Relation File Format): plain-text format with inline schema
 - .parquet: efficient binary columnar storage (preferred for large data)
- **Meta-data** annotations include:
 - Dataset description (source, intended use, licensing)
 - Feature names and types (numeric, categorical, ordinal, date, string)
 - Designated roles: input features, target variable, ignored columns
- **Automatically extracted information:**
 - Meta-features (e.g., number of instances, number of features)
 - Statistical summaries (missing values, sparsity, imbalance)
 - Baseline performance of simple models (e.g., majority class classifier)

OPENML TASKS

- A **task** defines a concrete ML problem on a dataset, e.g., in supervised ML:
 - Target feature (to be predicted)
 - Resampling strategy (e.g., 10-fold CV, holdout)
 - Evaluation metric (e.g., accuracy, RMSE)
- Tasks ensure reproducibility by fixing all relevant components
- **Flows** (i.e., ML pipelines or learners) are executed on *tasks* and produce *runs*

OPEN QUESTION: WHAT DATASETS/TASKS TO USE?

- Hard to compare results across papers
- Benchmarking is often done on small set of datasets
 - Question about generalization to other datasets
 - Cherry picking or arbitrary selection
 - Different versions, different train-test setups
- Publication bias
 - Published papers report good results
 - Interesting to know WHEN an algorithm works (and when it doesn't)

OPENML BENCHMARKING SUITES

- **Benchmarking suite:** curated collection of standardized OpenML tasks
- Provide unified, shareable task definitions with fixed resampling and target
- Accessible via APIs in R, Python, and Java
- Enable reproducible and comparable experiments toward shared research goals

EXAMPLE I: OPENML-CC18 (CLASSIFICATION)

► BISCHL2021

Standardized benchmark suite for evaluating supervised classification algorithms.

- 72 classification tasks
- Medium size: 500–100,000 observations, ≤ 5000 features (post encoding)
- Contains missing values and categorical features
- Excludes:
 - Strong class imbalance
 - Time series or group structure
 - Sparse data and free-form text
- Follows objective and subjective selection criteria (see paper)

Extension of CC18 selection principles to regression tasks.

- Uses same structural filters as CC18
- Includes only datasets with numeric targets and ≥ 5 distinct values
- Excludes:
 - Datasets trivially solved by linear models
 - Datasets with ethical/legal concerns
 - Datasets restricted from public benchmarking

EXAMPLE III: AUTOML BENCHMARK SUITE (AMLB)

► GIJSBERS2024

Designed to evaluate modern AutoML systems on realistic and diverse problems.

- 71 classification tasks and 33 regression tasks
- Stricter inclusion criteria for difficulty and real-world complexity
- Covers multiple domains; avoids trivial, overly cleaned, or synthetic data
- Excludes:
 - Free-text features
 - Time dependencies or metadata leaks

WHY PARALLELIZATION MATTERS

Benchmark Scenario (AutoML Study)

- 71 datasets $\times$ 10-fold cross-validation
- 9 AutoML systems evaluated
- 1 hour per system-dataset-fold combination

⇒ Total runtime: **6390 hours** = **266+ days** sequentially

Conclusion: *Benchmarking at scale is infeasible without parallelization.*